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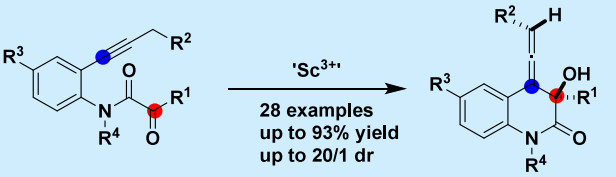
Sc(III)-Catalyzed Carbonyl−Yne Reaction for the Diastereoselective Allene Synthesis: A Concerted Asynchronous Mechanism

Xinya Yuan, Jiashen Yang, and Shaozhong Wang*

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ABSTRACT: The carbonyl−yne reaction, involving propargylic C−H bond cleavage for carbon−carbon bond formation, is significantly less developed than the classical Alder−ene reaction due to the linear geometry and relatively weak nucleophilicity of alkynes compared to alkenes. Herein, we report a $Sc(OTf)_3$-catalyzed carbonyl−yne reaction enabling the diastereoselective synthesis of multisubstituted allenes from α-ketoamide-tethered alkynes with 100% atom economy and excellent diastereoselectivity (up to 20/1 dr). DFT calculations support a concerted, asynchronous mechanism involving C−C bond formation and propargylic C−H cleavage. Theoretical analysis suggests that the observed stereoselectivity arises from an interplay of electronic and entropic effects in the transition states.

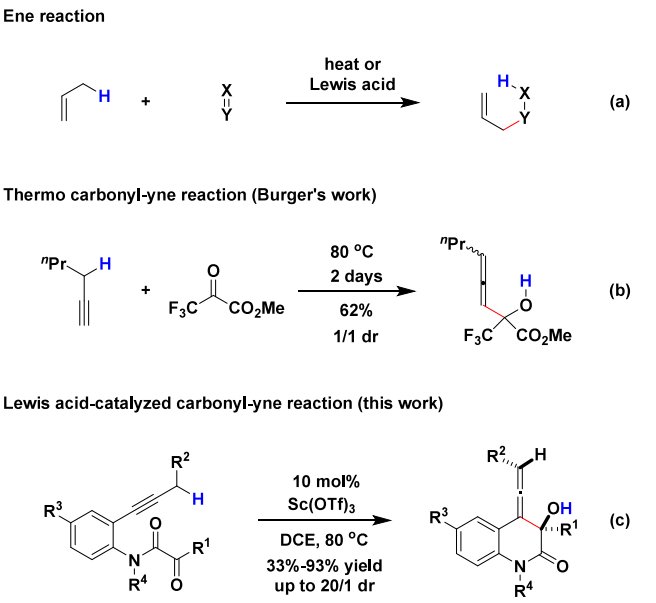


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Allenyl moieties constitute privileged axially chiral scaffolds with broad utility across organic synthesis, pharmaceuticals, and materials science.[1−4] Their orthogonal π-systems enable unique reactivity profiles, facilitating diverse transformations including cycloadditions, cycloisomerizations, allylic substitutions, and transition-metal-catalyzed couplings[5−10] While classical allene syntheses typically require stoichiometric reagents or multistep sequences,[11−14] catalytic carbonyl−yne reactions offer an atom-economical, convergent alternative for direct construction from alkynes and carbonyl partners. Mechanistically, this transformation parallels the classical ene reaction[15,16] but substitutes the "ene" component with a propargylic fragment (Scheme 1a,b). Where classical ene reactions engage an allylic C−H bond, an alkene C═C bond, and an enophile, the carbonyl−yne variant utilizes a propargylic C−H bond, an alkyne C≡C bond, and an ynophile (carbonyl group). Investigation of this "propargyl−ene-type" reaction would provide a rational mechanistic explanation and deeper understanding for the stereoselective formation of multisubstituted allene products. Nevertheless, achieving high diastereo- and even enantioselectivity for multisubstituted allenes, particularly sterically congested architectures, remains challenging.[17−20]

Existing carbonyl−yne reactions[21] can present certain limitations, such as a somewhat narrow substrate scope, modest levels of stereocontrol, and mechanistic aspects that are still being uncovered. Lewis acid catalysis might potentially overcome these limitations by coordinatively activating carbonyl groups toward alkyne nucleophiles.[22] Rare-earth metal triflates exhibit exceptional oxophilicity, functional group tolerance, and moisture compatibility, positioning them as ideal catalysts for the carbonyl−yne transformation.[23] Herein, we report the first $Sc(OTf)_3$-catalyzed carbonyl−yne reaction for diastereoselective synthesis of multisubstituted allenes from α-ketoamide-tethered alkynes (Scheme 1c). This method features perfect atom economy, broad substrate scope and exceptional diastereoselectivity. Mechanistic studies

Scheme 1. Reaction Design



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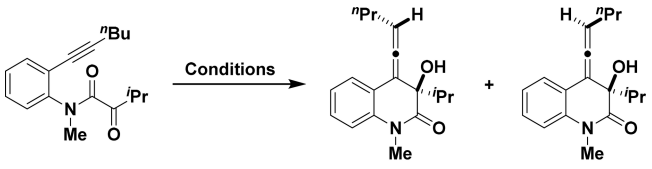
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supported by DFT calculations suggest a concerted asynchronous pathway where an interplay of electronic and entropic effects in the transition states might control the stereoselectivity. Furthermore, we demonstrate the synthetic versatility of the resulting allenes through diverse downstream transformations, including cycloisomerization and semipinacol rearrangements for accessing complex dihydrofurans.

We initiated our investigation using alkyne **1a** as a model substrate. Reactions conducted in DCE at room temperature with catalytic amounts of oxophilic Lewis acids including $Fe(OTf)_3$, $Cu(OTf)_2$, $Sn(OTf)_2$, and $Al(OTf)_3$ afforded allene **2a** in low yields with dr values ranging from 2.5/1 to 5/1 (Table 1, entries 1−4). In contrast, employing $In(OTf)_3$ as the catalyst gave the product in high yield with moderate dr (Table 1, entry 5). Subsequent screening of rare earth metal triflates [$Y(OTf)_3$, $La(OTf)_3$, $Sm(OTf)_3$, $Yb(OTf)_3$] yielded only trace amounts of **2a** (Table 1, entries 6−9). However, catalytic $Sc(OTf)_3$ provided the best diastereoselectivity, enabling the formation of **2a** in 45% yield (Table 1, entry 10). The relative stereochemistry was assigned based on X-ray of single crystal diffraction study of its analog **2u** (see SI for details). Using $Sc(OTf)_3$ as the catalyst, we observed similar results with DCM, toluene, or $PhCF_3$ as solvent (Table 1, entries 11−13). In contrast, the yield decreased significantly in dioxane (Table 1, entry14). Elevating the reaction temperature to 80 °C in 1,2-DCE improved the yield while maintaining comparable diastereoselectivity (Table 1, entries 15−16). Control experiments showed that reduced $Sc(OTf)_3$ loadings led to lower yields, and the carbonyl−yne reaction failed to occur either in the absence of $Sc(OTf)_3$ or in the presence of Brϕnsted acid HOTf, which could originate from the partial decomposition of $Sc(OTf)_3$ (Table 1, entries 17−19).

Table 1. Optimization for the Sc(III)-Catalyzed Carbonyl−yne Reaction[a]



1a 2a 2a'

Entry	Catalyst	Solvent	T (°C)	t (h)	Yield of 2a (%)[b]	dr[c]
1	$Fe(OTf)_3$	DCE	rt	12	11	2.5/1
2	$Cu(OTf)_2$	DCE	rt	12	17	3.4/1
3	$Sn(OTf)_2$	DCE	rt	12	36	2.6/1
4	$Al(OTf)_3$	DCE	rt	12	44	5/1
5	$In(OTf)_3$	DCE	rt	12	70	7/1
6	$Y(OTf)_3$	DCE	rt	12	trace	-
7	$La(OTf)_3$	DCE	rt	12	trace	-
8	$Sm(OTf)_3$	DCE	rt	12	9	6.8/1
9	$Yb(OTf)_3$	DCE	rt	12	12	9/1
10	$Sc(OTf)_3$	DCE	rt	12	45	10.5/1
11	$Sc(OTf)_3$	DCM	rt	12	45	9.7/1
12	$Sc(OTf)_3$	PhMe	rt	12	33	9/1
13	$Sc(OTf)_3$	$PhCF_3$	rt	12	32	9.3/1
14	$Sc(OTf)_3$	dioxane	rt	12	17	5.3/1
15	$Sc(OTf)_3$	DCE	60	6	61	8.5/1
16	$Sc(OTf)_3$	DCE	80	4	80	9/1
17[d]	$Sc(OTf)_3$	DCE	80	4	45	8.1/1
18[e]	-	DCE	80	12	0	-
19[f]	HOTf	DCE	80	4	0	-

[a]Reaction conditions: a mixture containing **1a** (0.1 mmol), Lewis acid (0.01 mmol), and solvent (1.0 mL) was stirred in a vial for the mentioned time. [b]NMR yield using mesitylene as an internal standard. [c]dr values were determined by the integration of peaks shown in ^1H NMR spectra. dr = diastereoisomeric ratio. [d]2.5 mol % of $Sc(OTf)_3$ was used. [e]In the absence of $Sc(OTf)_3$. [f]HOTf (0.03 mmol).

With optimized conditions established, we next explored the scope of the $Sc(OTf)_3$-catalyzed carbonyl−yne reaction by varying substituents on the ketone, alkyne, benzene ring, and nitrogen atom. Similar to substrate **1a**, alkynes **1b**−**1f** bearing aliphatic α-ketoamide groups with primary, secondary, or tertiary alkyl substituents (Me, Et, phenylethyl, Cy, *t*-Bu) underwent smooth conversion to allenes **2b**−**2f** in 33−88% yield with 5/1 to 8/1 dr (Table 2, entries 1−6). Alkynes **1g**−**1k** with arylated α-ketoamides reacted similarly, affording allenes **2g**−**2k** in 42−73% yield with 2.6/1 to 5.8/1 dr (Table 2, entries 7−11). Alkynes **1l**−**1u** with primary, secondary, tertiary alkyl, or phenyl substituents at the propargylic position (Et, phenethyl, *i*-Bu, cyanoethyl, $BnOCH_2$, 1-chloropropyl, 2-(1*H*-indol-1-yl)ethyl, *i*-Pr, cyclohexyl, Ph) all produced allenes **2l**−**2u** in excellent yields (up to 90%) with dr values up to 20/1 (Table 2, entries 12−21). Substrates **1v**−**1z** with substituents (Br, F, NO_2, Me, Ph) on the aniline ring linking the α-ketoamide and alkyne units also performed well, delivering allenes **2v**−**2z** in comparable efficiency (Table 2, entries 22−26). Finally, alkynes **1aa**−**1ab** featuring benzyl or propargyl groups on the nitrogen atom participated effectively, yielding allenes **2aa** (84% yield, 8/1 dr) and **2ab** (67% yield, 5.6/1 dr), respectively (Table 2, entries 27−28). However, in the case of alkynes **1ac**−**1ae** bearing secondary alkyl substituents (*i*-Pr, cyclohexyl, cyclopentyl) at the alkyne terminus, a Sc(III)-catalyzed tandem carbonyl−yne/cycloisomerization reaction occurred, furnishing polycyclic 2,5-dihydrofuran products **2ac**−**2ae** (Table 2, entries 29−30).

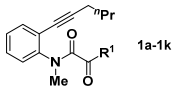
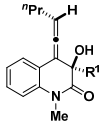
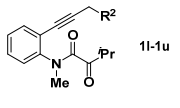
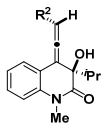
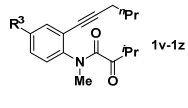
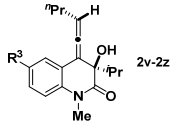
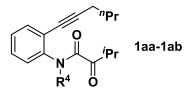
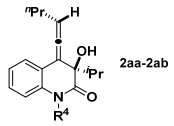
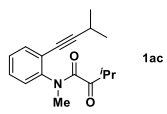
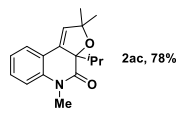
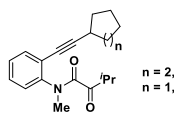
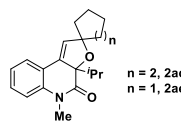
To elucidate the origin of diastereoselectivity in the scandium-catalyzed formation of multisubstituted allenes, density functional theory (DFT) calculations were performed (Figure 1).[24−38] The reaction initiates with coordination of the α-ketoamide group to scandium trifluoromethanesulfonate, forming intermediates **INT1** and **INT2**. These intermediates then undergo C−C bond formation and hydrogen migration, giving **INT3** and **INT4** via transition states **TS1** and **TS2**, respectively. Subsequent decomplexation yields allenes **2a** and **2a′**. The calculated Gibbs free energy of activation ($\Delta G^{\ddagger}$) for the pathway via **TS1** (8.7 kcal/mol) is 1.8 kcal/mol lower than that for the **TS2** pathway (10.5 kcal/mol), which is in excellent agreement with the experimentally observed product distribution. Geometric analysis of **TS1** indicates a concerted yet asynchronous mechanism.

This asynchronicity is evidenced by the bond lengths within the transition state structure: the forming C1−C2 bond (1.98 Å) is substantially formed, whereas the scissile C3−H bond is marginally elongated to 1.12 from 1.09 Å, and the H−O bond is only 2.16 Å. This disparity demonstrates that C−C bond formation is significantly more advanced than C−H bond cleavage.

A notable finding is that the overall free energy trend is inverted relative to the electronic energies. **TS2** is electronically more stable than **TS1** by approximately 0.19 kcal/mol. However, the diastereoselectivity is governed by the Gibbs free energy, and the preference for the **TS1** pathway is determined

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Table 2. Scope of Substrates[a]

Entry	Alkyne	Allene	Yield (%)[b]	Dr[c]
	 1a-1k	 2a-2k		
1	R^1 = ^{i}Pr, **1a**	**2a**	76	9/1
2	R^1 = Cyclohexyl, **1b**	**2b**	78	8/1
3	R^1 = Me, **1c**	**2c**	40	5/1
4	R^1 = Et, **1d**	**2d**	88	5.7/1
5	R^1 = $PhCH_2CH_2$, **1e**	**2e**	75	5/1
6	R^1 = ^{t}Bu, **1f**	**2f**	33	5.2/1
7	R^1 = Ph, **1g**	**2g**	42	5.8/1
8	R^1 = 4-ClC_6H_4, **1h**	**2h**	68	4.6/1
9	R^1 = 4-$CF_3C_6H_4$, **1i**	**2i**	73	5/1
10	R^1 = 4-$MeOC_6H_4$, **1j**	**2j**	52	2.6/1
11	R^1 = 2-naphthyl, **1k**	**2k**	46	3/1
	 1l-1u	 2l-2u		
12	R^2 = Et, **1l**	**2l**	90	11/1
13	R^2 = $PhCH_2CH_2$, **1m**	**2m**	88	6.6/1
14	R^2 = ^{i}Bu, **1n**	**2n**	71	9/1
15	R^2 = $ClCH_2CH_2CH_2$, **1o**	**2o**	82	8.1/1
16	R^2 = $NCCH_2CH_2$, **1p**	**2p**	81	7/1
17	R^2 = $BnOCH_2$, **1q**	**2q**	76	7.7/1
18	R^2 =1H- IndoleCH_2CH_2, **1r**	**2r**	75	7.6/1
19	R^2 = ^{i}Pr , **1s**	**2s**	48	17/1
20	R^2 = Cyclohexyl, **1t**	**2t**	65	12/1
21	R^2 = Ph, **1u**	**2u**	85	20/1
	 1v-1z	 2v-2z		
22	R^3 = Br, **1v**	**2v**	87	5/1
23	R^3 = F, **1w**	**2w**	75	10/1
24	R^3 = NO_2, **1x**	**2x**	83	20/1
25	R^3 = Me, **1y**	**2y**	93	8/1
26	R^3 = Ph, **1z**	**2z**	78	8/1
	 1aa-1ab	 2aa-2ab		
27	R^4 = Bn, **1aa**	**2aa**	84	8/1
28	R^4 = 1-propargyl, **1ab**	**2ab**	67	5.6/1
29	 1ac	 2ac, 78%		
30	 n = 2, 1ad; n = 1, 1ae	 n = 2, 2ad, 81%; n = 1, 2ae, 80%		

[a]Reaction conditions: a mixture containing **1** (0.1 mmol), $Sc(OTf)_3$ (0.01 mmol), and 1,2-DCE (1.0 mL) was stirred in a vial at 80 °C for 4 h. [b]Isolated yields. [c]dr values were determined by the integration of peaks shown in ^{1}H NMR spectra.

Table 3. Comparative Analysis of Electronic and Thermodynamic Parameters for TS1 and TS2

	TS1	TS2	Difference
$E_{(Hartree)}$	−3588.90038	−3588.90069	0.0003106
G_{corr}	0.340408	0.343547	−0.003139
H_{corr}	0.481621	0.482035	−0.000414
$T^*S_{(Entropy)}$	0.141213	0.138488	0.002725

by entropic contributions (Table 3). The butyl substituent in **TS1** possesses greater conformational and vibrational freedom, which manifests as a larger population of low-frequency vibrational modes. These modes make a substantial contribution to the vibrational entropy, thereby lowering the overall Gibbs free energy of **TS1**. In contrast, in **TS2**, the motion of the butyl group is sterically restricted by interactions with the triflate anion. This restriction results in an increase in its vibrational wavenumbers by approximately 5−10 cm^{-1} relative to **TS1**, leading to a significant entropic penalty and rendering this pathway less favorable. Furthermore, the influence of the R^1 (Me, Et, *i*-Pr, *t*-Bu) substituent on the ketone group might also be linked to dispersion forces. Interactions between R^1 and the *n*-butyl on the alkyne group could potentially narrow the electronic energy difference between **TS1** and **TS2**, thereby affecting the reaction's diastereoselectivity.

To demonstrate the synthetic potential of the Sc-catalyzed carbonyl−yne reaction, we performed a series of transformations on allene **2a** (Figure 2). Treatment of **2a** with silver(I) triflate induced cyclization, affording dihydrofuran **3** in 76% yield. Alternatively, palladium-catalyzed cyclization/allylation of **2a** produced dihydrofuran **4**. Employing *N*-iodosuccinimide (NIS) provided dihydrofuran **5** in 70% yield. Notably, the formation of dihydrofurans **3**−**5** proceeded diastereoselectively, as confirmed by X-ray crystallographic analysis. Under the promotion of *N*-bromosuccinimide (NBS), **2a** underwent a semipinacol rearrangement featuring 1,2-migration of the isopropyl group. This transformation furnished trisubstituted alkene **6** as the major product, whose structure was unambiguously confirmed by X-ray crystallographic analysis.

In summary, we have developed a $Sc(OTf)_3$-catalyzed carbonyl−yne reaction for the diastereoselective synthesis of multisubstituted allenes from *α*-ketoamide-tethered alkynes. This method features excellent atom economy, a broad substrate scope (tolerating diverse alkyl, aryl, and functional groups), and good-to-excellent diastereoselectivity. Theoretical analysis suggests that the observed stereoselectivity arises from a sophisticated interplay of electronic and entropic effects in the transition states. Meanwhile, this work addresses long-standing challenges in carbonyl−yne chemistry, establishing a robust platform for allene synthesis with applications in target-oriented synthesis. Further investigations into the asymmetric

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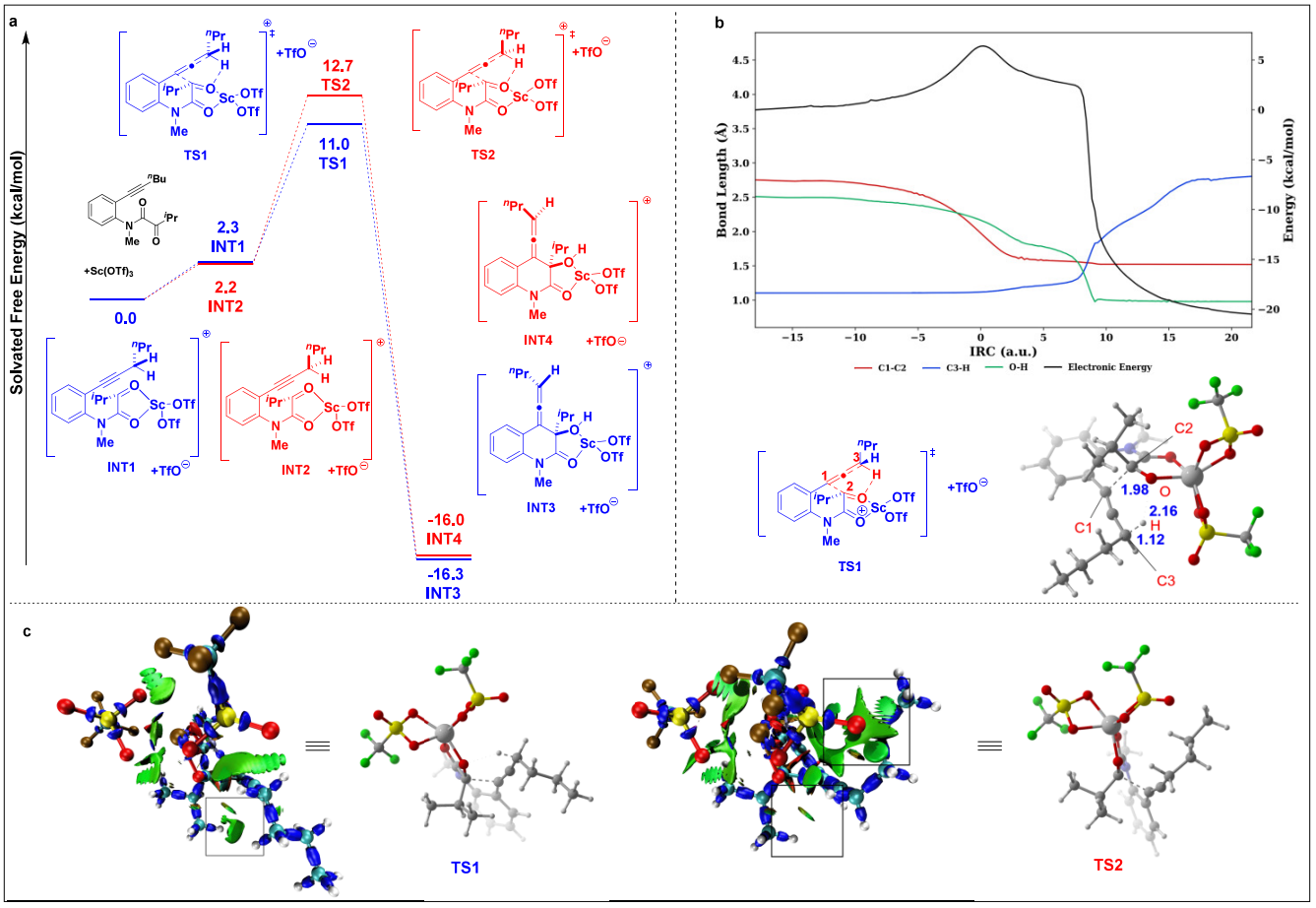


Figure 1. a) Potential energy profile of the Sc(III)-catalyzed carbonyl−yne reaction of **1a**. b) Solvated electronic energies and selected bond lengths along the IRC of **TS1**. Bond lengths labeled in **TS1** structure are in Å. c) IRI analysis of transition states **TS1** (top) and **TS2** (bottom). Areas emphasized by the squares are dispersion attraction and steric hindrance between isopropyl and butyl in **TS1**, dispersion attraction between triflate and butyl, and steric hindrance between isopropyl and butyl in **TS2**, respectively. The color scale represents: blue for strong attraction, green for van der Waals interactions, and red for strong repulsion.

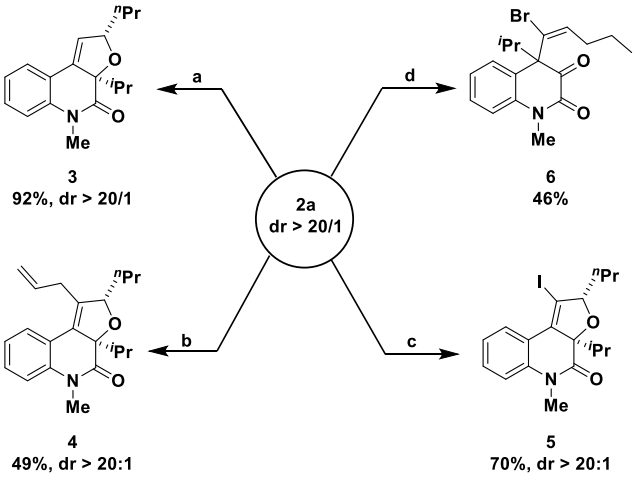


Figure 2. Synthetic applications of allene **2a**. a) AgOTf (0.1 equiv), acetone/H_2O (1/0.1 mL), rt, 12 h; b) $PdCl_2$ (20 mol %), CH_2Cl_2 (1.0 mL), allyl bromide (5.0 equiv), N_2, rt, 4 h; c) NIS (1.2 equiv), $CHCl_3$ (1.0 mL), rt, 3 h; d) NBS (1.2 equiv), $CHCl_3$ (1.0 mL), rt, 3 h.

carbonyl−yne reaction and its synthetic applications are underway in our laboratory.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c03172.

Experimental procedures, characterizations, and copies of ^{1}H and ^{13}C NMR spectra of products (PDF)

Accession Codes

Deposition Numbers 2391753, 2391994, 2403415, 2404896, 2404940, and 2483069 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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